

Friendly Order Type and Incomparability Components

A finite-case answer to a compositional question,
with transfinite extensions and executable certificates

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For every finite poset P ,

$$f(P) = |P| - c(\text{Inc}(P)).$$

Here f is Vialard's friendly order type and c counts all connected components of the incomparability graph, including isolated vertices. Consequently,

$$w(\mathcal{M}_{\text{DM}}(P)) = \omega^{|P| - c(\text{Inc}(P))}.$$

The report proves the component formula, classifies maximum friendly subsets, and gives composition laws, transfinite examples, and independently checked finite computations.

Scope and status. The source problem asks for a broader understanding and compositional calculus of friendly order type. This report resolves the finite specialization and a class of well-ordered sums; it does not resolve that entire program. The arguments are presented as proofs, but the manuscript has not been independently refereed or formally verified. A literature search did not locate the component formula; that is not a certification of priority.

Abstract

Friendly order type is the rank of the tree of bad sequences whose successive choices retain an incomparable witness in the current residual. It was introduced to compute the ordinal width of the multiset ordering. We prove that for a finite poset it is simply the rank of the graphic matroid of the incomparability graph: the number of vertices minus the number of components. The key lemma produces a maximal non-cut vertex while preserving any specified minimal vertex. It yields optimal sequences, spanning-forest certificates, and a classification of all maximum friendly subsets. Further consequences include closed formulas for finite Cartesian products and arbitrary nonempty finite lexicographic substitutions, an exact generating function for the distribution on naturally labelled posets, and a transfinite summation theorem. In particular, well-ordered sums of two-element antichains realize every ordinal as a friendly order type, even though all countably infinite examples have isomorphic incomparability graphs. Computations compare the defining recursion with the formula on 101,660 naturally labelled posets through size seven, and test additional product, substitution, and prescribed-root cases.

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1 The problem selected and the contribution

The question selected for this investigation is the compositional study of *friendly order type*. Vialard’s thesis explicitly asks whether this invariant can be calculated compositionally and whether it has a further structural interpretation [5, Conclusion, p. 109]. The finite specialization is particularly concrete: its definition is a recursive search over residual posets, but one can ask for a direct structural formula that avoids this search altogether.

Problem 1.1 (Finite specialization investigated here). Given an arbitrary finite poset P , identify a nonrecursive structural expression for its friendly order type. Use that expression to obtain composition laws and explicit optimal witnesses.

This is our specialization of the published research direction, not a claim that the literature separately labels this exact finite statement as a conjecture. There is already a substantial theory: the invariant and its connection to multiset width are due to Vialard, and the ordinal-sum law is also known [3, Definition 3.3, Theorem 3.4, Proposition 4.1(1)]. Those facts must not be represented as discoveries of this report.

The proposed contribution is a complete finite structural description, together with consequences proved below. The proof uses two ideas. First, incomparability components are not arbitrary pieces of a poset: they sit in a linearly ordered sequence. Second, in a finite connected incomparability graph one can remove a *maximal poset element* without disconnecting the graph. Repeating this operation leaves exactly one element per component. The defining sequence process can select every other element and no more.

The finite invariant therefore has a particularly simple graph-theoretic interpretation. Its transfinite analogue does not forget the ordering of the components; this distinction will be made explicit rather than hidden in an informal replacement of finite numbers by ordinals.

1.1 Literature and novelty audit

The audit used the published MFCS 2023 article, the author’s 2024 thesis, the earlier arXiv version, and the author’s publication page, accessed on 19 September 2026 [3, 5, 4, 2]. The earlier preprint uses the terminology *maximal safe order type*; this report uses the later tree-based definition of *friendly order type*. Exact-phrasal searches for that terminology and for an incomparability-component formula did not locate a later treatment of the formula proved here.

This audit establishes the provenance of the research question, but does not prove that the formula is unpublished. Its elementary ingredients may be known in other terminology. The mathematical assertions below have explicit proofs independent of any assertion of novelty. General infinite formulas not used in those proofs are deliberately not imported from earlier source versions.

2 Definitions, residuals, and rank conventions

We work in ZFC with set-sized orders. A *poset* is a set with a reflexive, antisymmetric, transitive relation $\leq$. We write $x \parallel y$ when neither $x \leq y$ nor $y \leq x$. The *incomparability graph* $\text{Inc}(P)$ has vertex set P and an undirected edge between distinct incomparable elements. Its number of connected components is denoted by $c(\text{Inc}(P))$; an isolated vertex counts as one component and the empty graph has zero components.

A finite sequence $s = (x_0, \dots, x_{k-1})$ is *bad* if $x_i \not\leq x_j$ whenever $i < j$. The residual after this sequence is

$$P_s = P \setminus \bigcup_{i < k} \uparrow x_i, \quad \uparrow x = \{y \in P : x \leq y\}.$$

Thus the next member of a bad sequence must lie in P_s . A poset is a *well-partial order*, or wpo, if it has no infinite bad sequence. Every finite poset is a wpo.

The tree of finite bad sequences, rooted at the empty sequence, is well-founded for a wpo. For a node s in any such tree, our rank convention is

$$\text{rk}(s) = \sup\{\text{rk}(sx) + 1 : sx \text{ is an immediate extension of } s\}, \quad \sup \emptyset = 0. \quad (1)$$

The rank of a tree means the rank of its root. In a finite tree this is the maximum number of edges on a root-to-leaf path. These are the standard rank conventions for ordinal invariants of wpos [1].

Definition 2.1 (Friendly order type). An *open-ended bad sequence* is obtained by repeatedly choosing $x \in P_s$ for which some $y \in P_s$ satisfies $x \parallel y$. The element y is a *friend* of that choice. The empty sequence is allowed. The friendly order type $f(P)$ is the rank of the tree of these sequences. This is the invariant denoted $o_{\perp}(P)$ by Vialard [3, Definition 3.3].

The friend is required in the residual *before* the choice is made. Since it is incomparable to the choice, it also survives that individual choice. It need not survive all later choices, and it need not eventually be selected. These distinctions matter: requiring a permanent friend or requiring a friend in the original poset would define different processes.

Write $P \perp x = \{y \in P : y \parallel x\}$ and $P \not\leq x = P \setminus \uparrow x$. Directly from the definition,

$$f(P) = \sup_{\substack{x \in P \\ P \perp x \neq \emptyset}} (f(P \not\leq x) + 1). \quad (2)$$

For finite P the supremum is a maximum. The next small observations also apply to infinite wpos.

Lemma 2.2. *If Q is an induced subposet of a wpo P , then $f(Q) \leq f(P)$. Moreover, $f(P) = 0$ exactly when P is a chain, including the empty chain. Every subsequence of an open-ended bad sequence is open-ended.*

Proof. A sequence using only elements of Q has the same comparisons in P . Each friend available inside Q is still available inside P , so its tree embeds as a subtree and ranks cannot increase on restriction. A first move exists precisely when some incomparable pair exists. Finally, deleting previous choices from a sequence only enlarges each subsequent residual. The friends witnessing the original sequence therefore witness any subsequence. $\square$

For a finite nonempty poset with n elements one immediately has

$$0 \leq f(P) \leq n - 1. \quad (3)$$

Indeed, all selected elements are distinct, and the last selected element needs a different element still present as a friend. Our task is to determine exactly how much stronger this bound becomes when the incomparability graph is disconnected.

3 The order structure of incomparability components

For posets A, B , their *ordinal sum* $A + B$ retains the internal orders and places every element of A below every element of B . An ordinal sum is very different from a disjoint union, which adds no cross comparisons.

Lemma 3.1 (Ordered components). *Distinct connected components of $\text{Inc}(P)$ are uniformly comparable: for two such components C, D , either every member of C is below every member of D , or conversely. These directions linearly order the components. If P is a wpo, this component order is a well-order.*

Proof. Elements in different components cannot be incomparable, since that would supply an edge joining the components. Suppose $a \parallel b$ and z lies outside their component. If $a < z$, then $z < b$ would imply $a < b$, a contradiction. Since b, z are comparable, necessarily $b < z$. The same argument handles the opposite direction.

Propagating along an incomparability path shows that the direction between z and all vertices of a component is constant. Propagating within the other component makes the direction uniform on both sides. Transitivity of the poset then gives transitivity of the component order, and distinct components are comparable, so it is linear.

If the component order of a wpo had an infinite strictly descending sequence, choosing one element from each component would give an infinite strictly descending sequence in P . This is impossible. In the usual set-theoretic setting, a linear order with no infinite descending sequence is a well-order. $\square$

In particular, every finite poset has a canonical decomposition

$$P = C_1 + \cdots + C_c, \quad (4)$$

where the C_i are its incomparability components in increasing order. This is a decomposition into *induced posets*; it does not assert that the components are chains or antichains.

Lemma 3.2 (Component capacity). *In any open-ended bad sequence in a finite poset, at most $|C| - 1$ elements can be selected from each incomparability component C .*

Proof. A friend of an element of C must also lie in C . If a sequence selected every element of C , consider its last choice from C . All other elements of C would already have been selected and hence removed. No friend could remain, a contradiction. Extra removals caused by upward closures cannot repair this obstruction. $\square$

Corollary 3.3. *For every finite poset,*

$$f(P) \leq |P| - c(\text{Inc}(P)).$$

The main issue is the lower bound: can these local capacities all be reached simultaneously? A generic graph deletion order does not suffice. To avoid accidentally removing further elements, the next selected vertex should be maximal in the remaining poset. The following lemma supplies exactly this.

4 A maximal non-cut lemma and the finite formula

A vertex is called *non-cut* here if deleting it leaves a connected graph. We apply this only to connected graphs with at least two vertices; the one-vertex graph left after a deletion is connected.

Lemma 4.1 (Maximal non-cut vertex with a protected root). *Let P be a finite poset with at least two elements and connected incomparability graph. Given any minimal element r of P , there exists $x \neq r$ such that x is maximal in P and $\text{Inc}(P \setminus \{x\})$ is connected.*

Proof. Choose a maximal element $x \neq r$. Such an element exists: if the only maximal element were r , finiteness would imply that every element is below r , contrary to the minimality of r and $|P| \geq 2$. If deleting x leaves a connected graph, we are done.

Otherwise let the components of $\text{Inc}(P \setminus \{x\})$ be

$$D_1 < D_2 < \cdots < D_k, \quad k \geq 2,$$

in the order provided by lemma 3.1. Each D_i has a vertex adjacent to x in the original incomparability graph. Otherwise D_i would have no edge either to x or to any other D_j , contradicting connectedness of the original graph.

Choose $z \in D_1$ with $z \parallel x$. For any $i \geq 2$ and $y \in D_i$, we have $z < y$. If $y < x$, transitivity would give $z < x$, a contradiction. If $x < y$, maximality of x would be contradicted. Hence

$$x \parallel y \quad \text{for every } y \in D_2 \cup \cdots \cup D_k. \quad (5)$$

Choose a maximal element y of the last component D_k . It is maximal in all of P : any element in an earlier component is below it, no element of D_k is above it, and it is incomparable with x . Also $y \neq r$, because any element of D_1 is below y , while r is minimal.

Deleting y leaves a connected graph. Indeed, D_1 remains connected and has an edge to x . Every remaining vertex of D_i for $i \geq 2$ is directly adjacent to x by (5), regardless of whether deletion of y disconnects the graph inside D_k . Thus y is the required vertex when the original choice x is a cut vertex. $\square$

Proposition 4.2 (Connected case). *If P is finite and $\text{Inc}(P)$ is connected, then*

$$f(P) = |P| - 1.$$

More strongly, for every minimal $r \in P$ there is an open-ended sequence selecting precisely $P \setminus \{r\}$. At each step it selects a maximal vertex of the current induced poset and leaves that poset connected.

Proof. A singleton gives the empty sequence. For at least two elements, apply lemma 4.1 with protected root r . The chosen x has a friend since the graph is connected and has more than one vertex. Since x is maximal, its upward closure in the current poset is just $\{x\}$; hence the residual is exactly $P \setminus \{x\}$, with a connected incomparability graph. The root r is still minimal there. Induction repeats this construction until only r remains. The sequence has $|P| - 1$ choices, and (3) proves optimality. $\square$

Theorem 4.3 (Finite component formula). *For every finite poset P , including the empty poset,*

$$\boxed{f(P) = |P| - c(\text{Inc}(P))}. \quad (6)$$

Proof. The empty case is immediate. In the decomposition $P = C_1 + \dots + C_c$, choose a minimal root r_i in each induced component C_i . By [proposition 4.2](#), there is a local open-ended sequence of length $|C_i| - 1$ leaving only r_i in that component.

Run these local sequences in the order $C_c, C_{c-1}, \dots, C_1$. A choice in one component cannot remove an element of an earlier component, so a component is still intact when its local sequence starts. A choice may remove surviving roots in later components, but those roots are not needed as friends: all local friends lie in the component currently being processed. Thus the concatenated sequence is legal and has length

$$\sum_{i=1}^c (|C_i| - 1) = |P| - c.$$

The upper bound is [corollary 3.3](#). □

Remark 4.4 (Why the proof is not just a graph argument). For arbitrary graphs one can repeatedly delete non-cut vertices. The extra requirement here is maximality in a compatible poset. Without it, deleting an upward closure could remove several vertices at once and destroy the intended sequence length. [Lemma 4.1](#) is the point where transitivity of the order supplies more structure than graph connectivity alone.

5 All maximum friendly subsets and forest certificates

For a finite subset $S \subseteq P$, say that S is *friendly in P* if it admits a linear extension whose reverse listing is an open-ended bad sequence in P . By [lemma 2.2](#), all subsequences of that listing are then open-ended as well. Conversely, an open-ended sequence listing all elements of S has a reverse listing that is a linear extension of S . This finite formulation agrees with the friendly-subset condition used in Vialard's alternative characterization [[3](#), Definition 4.4].

Theorem 5.1 (Classification of maximum friendly subsets). *Let $C_1 < \dots < C_c$ be the incomparability components of a finite poset P . The friendly subsets of maximum possible size are exactly*

$$P \setminus \{r_1, \dots, r_c\}, \quad r_i \in \text{Min}(C_i). \quad (7)$$

Their common size is $|P| - c$, and their number is

$$\prod_{i=1}^c |\text{Min}(C_i)|, \quad (8)$$

with empty product equal to 1.

Proof. The construction in [theorem 4.3](#) realizes every choice of roots in (7), proving sufficiency and the asserted size. For necessity, a maximum friendly subset must omit at least one member of each component by [lemma 3.2](#). Since its size is $|P| - c$, it omits exactly one, say r_i , from each C_i .

Fix a sequence listing the subset, and consider a component C_i . If r_i were not minimal in C_i , some selected element $a \in C_i$ would satisfy $a < r_i$. At the choice of a , the root r_i would be removed.

If this were not the last choice from C_i , then no unselected member of C_i could remain as a friend for the last choice. If it were the last choice, the only possibly remaining other member of C_i would be r_i , which is comparable to a . Both possibilities violate the friend condition. Thus each r_i is minimal. Different root choices give different complements, so multiplying the numbers of choices yields (8). $\square$

The theorem is stronger than merely computing a rank. It identifies every largest witness subset, although it does not enumerate all permissible orderings of a given witness subset.

Proposition 5.2 (Spanning-forest certificate). *The construction in [theorem 4.3](#) can attach to every choice x a friend y such that the edges $\{x, y\}$ form a spanning tree in each original incomparability component. The unselected r_i are the roots of those trees.*

Proof. Within a component, choose as friend any neighbor remaining after deleting the selected maximal vertex. Orient the witness edge from that vertex to its friend. The endpoint is either selected later in that local sequence or is its protected root. Following oriented edges therefore strictly increases deletion time until reaching the root; cycles are impossible. Every non-root has exactly one outgoing edge and eventually reaches the root, so the edges form a connected acyclic spanning graph. The construction never uses a friend from a different component, giving a spanning forest on the whole poset. $\square$

Corollary 5.3 (Graphic matroid rank and finite duality). *For finite P , $f(P)$ is the maximum number of edges in a forest of $\text{Inc}(P)$. In particular,*

$$f(P^{\text{op}}) = f(P).$$

Proof. A tree on m vertices has $m - 1$ edges, so a maximum forest in a finite graph has one spanning tree per component and $|P| - c$ edges. This is precisely the rank of its graphic matroid. Reversing the order does not change which pairs are incomparable, so it does not change this rank. $\square$

Not every spanning forest, and not every graph deletion order, is automatically a legal certificate. The theorem concerns the value of the graphic rank; [proposition 5.2](#) supplies particular forests compatible with the residual process.

6 Consequences for the ordinal width of multiset orders

A finite multiset on P is a finitely supported function $P \rightarrow \mathbb{N}$. After cancelling the common part of two multisets μ, ν , the strict multiset ordering $\mu <_{\text{DM}} \nu$ holds if the remainder of ν is nonempty and every element remaining in μ is strictly below some remaining element of ν . A witness in ν may be reused. We add equality to obtain $\leq_{\text{DM}}$. This is the multiset ordering, not the injective multiset embedding order [3, Section 1].

For a wpo Q , its *ordinal width* $w(Q)$ is the rank of the tree of finite sequences of pairwise incomparable elements, with convention (1). This agrees with ordinary antichain width on finite posets, but may be a transfinite ordinal on an infinite wpo.

Theorem 6.1 (Vialard’s width transfer, imported). *For every wpo P ,*

$$w(\mathcal{M}_{\text{DM}}(P)) = \omega^{f(P)}. \quad (9)$$

This is the existing multiset-width theorem [5, Theorem 5.3.3]; its technical proof is not reproduced or claimed as new here. All the finite computations of its exponent below follow from the preceding self-contained argument.

Corollary 6.2 (Closed multiset-width formula). *For every finite poset P ,*

$$w(\mathcal{M}_{\text{DM}}(P)) = \omega^{|P| - c(\text{Inc}(P))}.$$

Proof. Substitute [theorem 4.3](#) into (9). □

The case $P = \emptyset$ is worth keeping: there is exactly one multiset, the empty one, so the width is $1 = \omega^0$, not 0. For a nonempty finite chain the exponent is also zero, consistently with its multiset ordering being a chain.

Example 6.3 (The three-element distinction). Let $V = C_1 + A_2$ be a bottom element below two incomparable maxima, and let $D = C_2 \sqcup C_1$ be a two-element chain disjoint from a singleton. Here C_m denotes an m -element chain and A_m an m -element antichain. Both posets have cardinality and maximal order type 3, height 2, and ordinary width 2. But $\text{Inc}(V)$ has two components whereas $\text{Inc}(D)$ is connected. Therefore

$$f(V) = 1, \quad f(D) = 2, \quad w(\mathcal{M}_{\text{DM}}(V)) = \omega, \quad w(\mathcal{M}_{\text{DM}}(D)) = \omega^2.$$

Thus the failure of determination by cardinality, height, and ordinary width already occurs on three elements. On at most two elements, those data distinguish the possible isomorphism types, so this size is minimal.

7 Finite composition laws

All the formulas in this section apply to arbitrary finite posets, not only to posets generated from chains by a restricted list of operations.

7.1 Ordinal sums and disjoint unions

Corollary 7.1. *For finite A, B ,*

$$f(A + B) = f(A) + f(B).$$

If A, B are both nonempty, then

$$f(A \sqcup B) = |A| + |B| - 1.$$

Proof. The graph of an ordinal sum is the disjoint union of the two incomparability graphs, so both vertex and component counts add. In the poset disjoint union, every vertex of A is incomparable to every vertex of B . The resulting graph is connected when both are nonempty. Apply [theorem 4.3](#). □

The first identity is a finite instance of a known ordinal-sum law, and the finite disjoint-union identity is already consistent with the published finite formulas. The contribution here is their common structural explanation, not priority for these special cases.

7.2 Arbitrary lexicographic substitution

Let P be a finite base poset. For every $p \in P$, let Q_p be a nonempty finite poset. The *lexicographic substitution* $P[Q_p]$ consists of pairs (p, q) with $q \in Q_p$, ordered by

$$(p, q) \leq (p', q') \iff p < p' \text{ or } (p = p' \text{ and } q \leq q').$$

In particular, fibers over incomparable base elements are completely incomparable to one another.

Theorem 7.2 (Substitution formula). *Let C be the set of components of $\text{Inc}(P)$. Then*

$$f(P[Q_p]) = \sum_{\substack{C \in \mathcal{C} \\ |C| \geq 2}} \left(\sum_{p \in C} |Q_p| - 1 \right) + \sum_{\{p\} \in \mathcal{C}} f(Q_p). \quad (10)$$

Proof. There are no incomparability edges between fibers lying over different base components. Over a singleton component $\{p\}$, the incomparability graph is just $\text{Inc}(Q_p)$ and its contribution is $f(Q_p)$.

Now let C be a connected base component with at least two vertices. Each base vertex has a neighbor, and fibers are nonempty. Any two vertices in the same fiber can be joined by a path of length two through a neighboring fiber. A path in $\text{Inc}(P)$ connecting two base vertices lifts to a path connecting any prescribed vertices in their fibers, since adjacent fibers are joined by all possible edges. Thus the entire collection of fibers over C is one connected component, with $\sum_{p \in C} |Q_p|$ vertices. Its friendly contribution is that number minus one. Summing proves the formula. $\square$

Corollary 7.3 (Uniform fibers). *Suppose all fibers are the same nonempty finite poset Q of size m . If P has n elements, s singleton incomparability components, and t non-singleton incomparability components, then*

$$f(P[Q]) = m(n - s) - t + s f(Q).$$

Empty fibers cannot simply be substituted into (10): they may destroy paths in the base incomparability graph. When some fibers are empty, first delete their base vertices, take the induced base order, and apply the theorem to the remaining nonempty fibers.

7.3 Cartesian products

For a nonempty finite poset P , put

$$\varepsilon_{\perp}(P) = \begin{cases} 1 & \text{if } P \text{ has a least element,} \\ 0 & \text{otherwise,} \end{cases} \quad \varepsilon_{\top}(P) = \begin{cases} 1 & \text{if } P \text{ has a greatest element,} \\ 0 & \text{otherwise.} \end{cases}$$

These are global least and greatest elements, not merely minimal and maximal elements.

Theorem 7.4 (Cartesian-product formula). *Let A, B be finite posets with $|A|, |B| \geq 2$. Then*

$$f(A \times B) = |A||B| - 1 - \varepsilon_{\perp}(A)\varepsilon_{\perp}(B) - \varepsilon_{\top}(A)\varepsilon_{\top}(B). \quad (11)$$

Equivalently, the incomparability graph has exactly one non-singleton component, together with an isolated joint least element when both factors have least elements and an isolated joint greatest element when both have greatest elements.

Proof. Choose linear extensions of A and B , listing them as $a_0, \dots, a_{m-1}$ and $b_0, \dots, b_{n-1}$. Adding comparisons to make each list a chain can only delete incomparability edges from the product. Consequently the incomparability graph of the chain grid $C_m \times C_n$ is a spanning subgraph of $\text{Inc}(A \times B)$.

In that grid, all vertices except $(0, 0)$ and $(m-1, n-1)$ form a connected incomparability graph. To see this, use the two incomparable anchors

$$u = (0, n-1), \quad v = (m-1, 0).$$

A vertex (i, j) with $i > 0$ and $j < n-1$ is adjacent to u ; a vertex with $i < m-1$ and $j > 0$ is adjacent to v . Every non-corner vertex is an anchor or satisfies at least one of these conditions. The anchors are adjacent, so these vertices form one connected component in the original product too.

It remains to classify the two possible corner exceptions. The lower corner (a_0, b_0) is isolated if a_0, b_0 are both least elements, since it is then below every vertex. Otherwise, say a_0 is not least. Being first in a linear extension, it is minimal, so some $a \in A$ is incomparable with a_0 . Then (a, b_0) is incomparable with the lower corner. It is a non-corner grid vertex because $a \neq a_0$ and $b_0 \neq b_{n-1}$. Thus the lower corner joins the large component unless it is genuinely a least element. The argument for the upper corner is dual.

No other component can exist. The number of components is therefore $1 + \varepsilon_\perp(A)\varepsilon_\perp(B) + \varepsilon_\top(A)\varepsilon_\top(B)$, and [theorem 4.3](#) gives the formula. $\square$

If a factor is empty, the product is empty. If one factor is a singleton, the product is isomorphic to the other factor; these cases must not be put into (11) without its size hypotheses.

Corollary 7.5 (Several factors and Boolean lattices). *For finite $P_1, \dots, P_d$ with $d \geq 2$ and $|P_i| \geq 2$,*

$$f\left(\prod_{i=1}^d P_i\right) = \prod_{i=1}^d |P_i| - 1 - \prod_{i=1}^d \varepsilon_\perp(P_i) - \prod_{i=1}^d \varepsilon_\top(P_i).$$

In particular, for the Boolean lattice B_d with $d \geq 2$,

$$f(B_d) = 2^d - 3, \quad w(\mathcal{M}_{\text{DM}}(B_d)) = \omega^{2^d-3}.$$

Proof. Apply [theorem 7.4](#) successively; a product has a least, respectively greatest, element exactly when every factor does. Now use $B_d \cong C_2^d$ and [corollary 6.2](#). $\square$

8 Well-ordered sums and genuine transfinite values

The finite graph formula involves ordinary subtraction. There is no reason to replace it by an unspecified ordinal subtraction. The correct extension on well-ordered sums uses *ordinal addition*.

Lemma 8.1 (Binary ordinal sums). *For arbitrary wpos A, B ,*

$$f(A + B) = f(A) + f(B), \tag{12}$$

where the addition on the right is ordinal addition.

Proof. We give a residual proof of the known sum law. Fix A and argue by induction on the rank of $\text{Bad}(B)$. A move from A has friends only in A and leaves $A \not\geq x$, deleting all of B . A move from B has friends only in B and leaves $A + (B \not\geq x)$. The latter residual of B has smaller bad-sequence rank, so the induction hypothesis applies to it.

If $f(B) = 0$, there are no moves from B , and (2) gives $f(A + B) = f(A)$. Otherwise the contributions from moves in B have supremum

$$\sup_{x \text{ with a friend in } B} (f(A) + f(B \not\geq x) + 1) = f(A) + f(B).$$

Here associativity and continuity of ordinal addition in its right argument justify the equality; when the supremum is a successor, it is attained. Contributions from moves in A have supremum $f(A)$ and cannot enlarge this value. This proves (12). $\square$

Theorem 8.2 (Transfinite sum law). *Let α be an ordinal and let $(P_\xi)_{\xi < \alpha}$ be wpos. Their well-ordered sum is a wpo and*

$$f\left(\sum_{\xi < \alpha} P_\xi\right) = \sum_{\xi < \alpha} f(P_\xi). \quad (13)$$

Proof. In a bad sequence, the indices of its successive terms must be nonincreasing: a term in a strictly earlier block is below every term in a later block. An infinite nonincreasing sequence of ordinals is eventually constant, so an infinite bad sequence in the sum would eventually give an infinite bad sequence inside one block. Hence the sum is a wpo.

Prove the formula by transfinite induction on α . The zero case is immediate and successor stages are lemma 8.1. Let α be a nonzero limit. Every nonempty open-ended sequence starts in some block P_ξ , and all its later choices lie in blocks of index at most ξ . Every friend of a choice is in the same block as that choice. Thus the entire subtree below this first choice is unchanged if the whole order is replaced by its prefix through P_ξ .

Applying the rank recursion at the root therefore gives

$$f\left(\sum_{\xi < \alpha} P_\xi\right) = \sup_{\beta < \alpha} f\left(\sum_{\xi < \beta} P_\xi\right).$$

The induction hypothesis and the definition of a limit ordinal sum turn the right-hand side into $\sum_{\xi < \alpha} f(P_\xi)$. $\square$

Corollary 8.3 (Finite blocks). *For any well-ordered family of finite posets,*

$$f\left(\sum_{\xi < \alpha} P_\xi\right) = \sum_{\xi < \alpha} (|P_\xi| - c(\text{Inc}(P_\xi))). \quad (14)$$

In particular, if every incomparability component of a wpo P is finite and the components are $(C_\xi)_{\xi < \alpha}$ in their canonical order, then

$$f(P) = \sum_{\xi < \alpha} (|C_\xi| - 1). \quad (15)$$

Proof. Use theorem 4.3 in theorem 8.2. For the second statement, lemma 3.1 provides the well-ordered component decomposition. $\square$

The finite subtraction takes place *inside each summand*; the outer sum is ordinal addition in the specified order. Consequently a final finite block can survive in the answer when the same block at the beginning would be absorbed.

Example 8.4 (Explicit ordinal widths). Let $H = \sum_{n < \omega} A_2$. Then

$$f(H) = \omega, \quad w(\mathcal{M}_{\text{DM}}(H)) = \omega^\omega.$$

For $k < \omega$,

$$f(H + A_{k+1}) = \omega + k, \quad w(\mathcal{M}_{\text{DM}}(H + A_{k+1})) = \omega^{\omega+k}.$$

Likewise, $\sum_{\xi < \omega^2} A_2$ has friendly order type ω^2 and multiset ordinal width ω^{ω^2} . Inserting arbitrary chain blocks does not add to the friendly order type, since their contribution is zero.

Corollary 8.5 (Every ordinal at ordinary width two). *For every ordinal α , put*

$$P_\alpha = \sum_{\xi < \alpha} A_2.$$

Then $f(P_\alpha) = \alpha$ and $w(\mathcal{M}_{\text{DM}}(P_\alpha)) = \omega^\alpha$. For $\alpha > 0$, the ordinary maximum antichain size of P_α is exactly two.

Proof. Every two-element antichain contributes one to (13). Antichains cannot meet two different blocks, and each nonempty block has an antichain of size two. Apply the width-transfer theorem. $\square$

Proposition 8.6 (The infinite graph-only obstruction). *On infinite wpos, friendly order type is not determined by the isomorphism type of the incomparability graph, even if every component is finite. In fact, for all countably infinite ordinals α , the graphs $\text{Inc}(P_\alpha)$ are isomorphic, while $f(P_\alpha) = \alpha$.*

Proof. Each graph is a disjoint union of countably infinitely many isolated edges. Any bijection between their block index sets induces a graph isomorphism. The friendly order types are given by corollary 8.5 and need not coincide; for example, ω and $\omega + 1$ give different answers. $\square$

Thus the finite graphic-rank interpretation is exact but does not extend unchanged to infinite graphs. The component order supplies indispensable ordinal information. The transfinite examples above are consequences of the sum law, not claims that the general connected infinite case has been solved.

9 Sharp finite bounds and loss under added comparisons

Let $\text{str}(P)$ be the induced subposet obtained by deleting all vertices that are comparable with every element of P . These are exactly the isolated vertices of $\text{Inc}(P)$.

Proposition 9.1 (Exact stripped formula and sharp bounds). *Let $s = |\text{str}(P)|$. Then $f(P) = f(\text{str}(P))$. If $s = 0$, the value is zero. Otherwise $s \geq 2$, and if t is the number of non-singleton incomparability components,*

$$f(P) = s - t, \quad \left\lceil \frac{s}{2} \right\rceil \leq f(P) \leq s - 1.$$

For each fixed $s \geq 2$, every integer in this interval is attained.

Proof. Deleting an isolated vertex reduces the vertex count and the component count by one, leaving their difference unchanged. The t remaining components have at least two vertices each, so $1 \leq t \leq \lfloor s/2 \rfloor$. This proves the bounds. Conversely, for any such t , partition s into t positive parts each at least two and take the ordinal sum of antichains of those sizes. Its non-singleton component count is t . $\square$

For an n -element poset with no restriction on isolated vertices, every integer from 0 through $n - 1$ is possible: to realize k , use $C_{n-k-1} + A_{k+1}$, with the empty-chain convention when necessary.

Proposition 9.2 (Maximal separation at fixed elementary invariants). *For every $n \geq 3$, the posets*

$$L_n = C_{n-2} + A_2, \quad R_n = C_{n-1} \sqcup C_1$$

both have cardinality and maximal order type n , height $n - 1$, and ordinary width 2, but

$$f(L_n) = 1, \quad f(R_n) = n - 1, \quad w(\mathcal{M}_{\text{DM}}(L_n)) = \omega, \quad w(\mathcal{M}_{\text{DM}}(R_n)) = \omega^{n-1}.$$

The friendly-order-type gap $n - 2$ is the largest possible between two non-chain n -element posets.

Proof. The chain summand of L_n contributes $n - 2$ isolated vertices and its antichain contributes one component; hence $f(L_n) = 1$. The singleton in the disjoint union R_n is incomparable to all $n - 1$ chain elements, so its incomparability graph is connected and $f(R_n) = n - 1$. The height and width assertions are immediate from the two constructions. Every non-chain finite poset has friendly order type at least one by [lemma 2.2](#) and at most $n - 1$ by (3); hence the gap is maximal. $\square$

Proposition 9.3 (Exact monotonicity under augmentation). *Suppose P and Q are finite posets on the same support and every comparison of P is also a comparison of Q . Then*

$$f(P) - f(Q) = c(\text{Inc}(Q)) - c(\text{Inc}(P)) \geq 0.$$

Proof. Adding comparisons removes incomparability edges. Removing edges can only split components, not join them. The equality is obtained by subtracting the two instances of [theorem 4.3](#). $\square$

This statement applies to the completed transitive order. Adding one proposed comparison can force many others by transitivity, so it need not lower the invariant by only one.

10 Enumeration and a bivariate generating function

Call a poset on $\{1, \dots, n\}$ *naturally labelled* when $i <_P j$ implies $i < j$ as integers. Let a_n count such posets, with $a_0 = 1$, and let b_n count the ones with connected incomparability graph for $n \geq 1$. These are not counts of unlabelled posets or of all labelled posets. Every isomorphism type has a natural labelling, but may have many of them.

Put

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad B(z) = \sum_{n \geq 1} b_n z^n.$$

These are formal power series; no analytic convergence is asserted. Let $a_{n,k}$ count naturally labelled n -element posets with friendly order type k .

Theorem 10.1 (Distribution by component rank). *One has*

$$A(z) = \frac{1}{1 - B(z)}, \quad b_n = a_n - \sum_{i=1}^{n-1} b_i a_{n-i}, \quad (16)$$

and the bivariate distribution is

$$\sum_{n,k \geq 0} a_{n,k} z^n u^k = \frac{1}{1 - \sum_{m \geq 1} b_m z^m u^{m-1}}. \quad (17)$$

For $n \geq 1$ and $0 \leq k \leq n-1$,

$$a_{n,k} = [z^n] B(z)^{n-k}. \quad (18)$$

Proof. In a natural labelling, if $C < D$ are incomparability components, every label in C is smaller than every label in D . Therefore components occupy consecutive intervals of labels in their canonical order. The component decomposition is consequently a unique ordered list of nonempty naturally labelled connected-incomparability posets. Conversely, concatenating any such list by ordinal sum gives a naturally labelled poset with exactly those components. Summing over ordered lists gives $1 + B + B^2 + \dots$, proving the first identity. Extracting its coefficients proves the recurrence.

A component of size m contributes $m-1$ to friendly order type by [theorem 4.3](#). Weighting it by $z^m u^{m-1}$ gives (17). A poset of size n and friendly order type k has exactly $n-k$ components, so extracting from lists of that length gives (18). $\square$

The exhaustive counts give

$$(a_0, \dots, a_7) = (1, 1, 2, 7, 40, 357, 4824, 96428)$$

and

$$(b_1, \dots, b_7) = (1, 1, 4, 27, 275, 4070, 86278).$$

These values in this report are outputs of the included exact enumeration, not numbers taken from a database without verification.

Corollary 10.2 (Fixed-rank counting polynomials). *For fixed $k \geq 0$ and $n \geq k+1$,*

$$a_{n,k} = \sum_{\substack{m_1, \dots, m_k \geq 0 \\ \sum_{j=1}^k j m_j = k}} \frac{(n-k)_r}{\prod_{j=1}^k m_j!} \prod_{j=1}^k b_{j+1}^{m_j}, \quad r = \sum_{j=1}^k m_j, \quad (19)$$

where $(x)_r = x(x-1)\dots(x-r+1)$ and empty products are 1. It is a polynomial in n of degree k with leading coefficient $1/k!$.

Proof. There are $n-k$ component positions. Let m_j record how many components have size $j+1$. Their total rank contribution is $\sum j m_j = k$, while all remaining components are singletons. Choose and order the r non-singleton positions in $(n-k)_r / \prod m_j!$ ways, then choose the connected structures on each position. This gives the formula. The degree is at most $r \leq k$. Degree k occurs only when $m_1 = k$; since $b_2 = 1$, that term is $\binom{n-k}{k}$ and has leading coefficient $1/k!$. $\square$

For example, in their indicated range of n ,

$$\begin{aligned} a_{n,0} &= 1, \\ a_{n,1} &= n - 1, \\ a_{n,2} &= 4(n - 2) + \binom{n-2}{2} = \frac{(n-2)(n+5)}{2}, \\ a_{n,3} &= 27(n - 3) + 4(n - 3)(n - 4) + \binom{n-3}{3}. \end{aligned}$$

These formulas offer a separate combinatorial explanation for the initial columns of the computation table, rather than only an empirical pattern.

11 Algorithms and certificate verification

The package implements two intentionally different ways to calculate the finite invariant. The structural algorithm constructs the incomparability graph, counts connected components, and returns $n - c$. The reference algorithm uses only the defining residual recursion.

11.1 Direct computation of the value

With a complete $n \times n$ comparison matrix, incomparability testing and graph construction take $O(n^2)$ Boolean operations. Connected components can be found in $O(n^2)$ time in this dense representation. Thus the friendly order type and the finite exponent in the multiset-width formula are computed in $O(n^2)$ time from a valid comparison matrix. The returned ordinal is represented symbolically as ω^k ; there is no attempt to enumerate the infinite multiset order.

Input validation is separate. A naive transitivity check takes $O(n^3)$ Boolean operations, and input consisting only of generating strict relations requires transitive closure first. The supplied command-line interface accepts such relations, computes their closure, and rejects cycles. The quadratic claim is for evaluation after a valid comparison matrix is given, not for all possible input representations.

11.2 Constructing an optimal witness

Decompose the graph into its ordered components. Process them in decreasing component order. Within a component, repeatedly apply [lemma 4.1](#), record an incomparable neighbor as friend, and delete the selected maximal vertex. Stop when only the chosen root remains. A root can be specified in advance, provided it is minimal in that component.

An implementation of one non-cut step first selects a maximal vertex, computes the components after its deletion, and either uses that vertex or selects a maximal element of the last new component as in the proof. Each such step needs $O(n^2)$ elementary operations on a comparison matrix, so the full witness can be constructed in $O(n^3)$ time. It contains at most $n - 1$ choice–friend pairs. The Python implementation uses integer bit sets; these matrix bounds describe the elementary-operation model, rather than pretending that arbitrarily long machine integers have unit cost.

A verifier simulates the residuals and checks that each named choice and friend is still present and incomparable. It also checks that the witness edges are acyclic and span each original component.

This establishes legality and checks the claimed forest certificate. Optimality is then certified by comparing the sequence length with the component upper bound.

11.3 Independent residual recursion

For a subset mask R , define

$$F(R) = \max(\{0\} \cup \{1 + F(R \setminus \uparrow x) : x \in R, R \cap (P \perp x) \neq \emptyset\}). \quad (20)$$

Memoizing this recurrence considers at most 2^n subsets, with at most n choices at each subset. Its worst-case count is $O(n2^n)$ subset transitions, before accounting for the cost of operations on the bit sets. It does not call the component formula and does not use the maximal non-cut lemma. Thus comparison against it is not merely testing the formula against itself.

11.4 Using the supplied program

The example input `examples/diamond.json` is

```
{"n":4,"relations":[[0,1],[0,2],[1,3],[2,3]]}
```

Run the following from the archive directory:

```
python src/friendly.py examples/diamond.json --dp
python src/verify.py --max-n 7
```

The first command reports components $\{0\}, \{1, 2\}, \{3\}$, friendly order type 1, multiset ordinal width ω^1 , and an independently checked certificate. Input labels are integers from 0 through $n - 1$; they need not themselves form a linear extension. Optional input `"roots": [...]` prescribes one minimal root per component. The code uses Python 3.10 or later and only the standard library.

12 Exact computational checks

The delivered run was performed with Python 3.13.5. It generated every naturally labelled poset through size seven. To pass from size $n - 1$ to size n , the strict predecessor set of the new largest label is chosen among the order ideals of the previous poset. Every such ideal gives a valid natural order, and every natural order arises once in this way. Since every finite poset admits a linear extension, all isomorphism types through the bound are represented, generally with multiplicity.

For each generated order, the test compares (20) with the graph formula, repeats the direct recursion for the dual order, constructs an optimal certificate, and simulates it. The maximal non-cut conclusion is additionally checked at the root of every connected case of size at least two.

n	Total	$f = 0$	$f = 1$	$f = 2$	$f = 3$	$f = 4$	$f = 5$	$f = 6$
0	1	1	–	–	–	–	–	–
1	1	1	–	–	–	–	–	–
2	2	1	1	–	–	–	–	–
3	7	1	2	4	–	–	–	–
4	40	1	3	9	27	–	–	–
5	357	1	4	15	62	275	–	–
6	4,824	1	5	22	106	620	4,070	–
7	96,428	1	6	30	160	1,047	8,906	86,278

The total is 101,660 orders, with 90,655 explicit connected-case non-cut checks. Additional tests cover all 2,601 ordered pairs of naturally labelled factors of sizes zero through four, comparing both graph counts and the independent residual recursion with the Cartesian-product formula. The largest product supports have sixteen elements.

For substitution, every natural base of size at most four is tested with each fiber independently selected from a singleton, a two-element chain, a two-element antichain, and a three-element chain. This gives 10,725 heterogeneous substitutions, each checked against direct residual recursion. For maximum friendly subsets, 2,365 candidate subsets of the predicted maximum size are tested through support size five by a recursion restricting choices to that subset. Exactly the subsets in [theorem 5.1](#) pass; 1,052 corresponding prescribed-root certificates are constructed and verified. Eight malformed-order or malformed-friend cases are rejected as expected.

All tests passed. The full run recorded 15.584 seconds in this environment; this is a reproducibility datum, not a performance guarantee. The JSON report and CSV rank distribution are included under data/.

The generator is exhaustive only in the stated finite ranges. It does not verify a theorem for all finite posets, and it does not verify a transfinite rank computation. The proofs above supply those assertions. The computation is useful for detecting definition mistakes, counterexamples, off-by-one errors, incorrect empty cases, and errors in certificate construction. No Lean formalization is claimed.

13 What has been settled, and what has not

The finite specialization has an exact answer with a complete proof:

$$f(P) = |P| - c(\text{Inc}(P)).$$

It yields all maximum friendly subsets, efficient certificates, finite substitution and product laws, sharp bounds, and an enumerative distribution. Together with a sum law, it also evaluates all wpos whose incomparability components are finite.

The broader problem from the literature remains larger than these statements. A wpo may have infinite incomparability components. The connected finite proof relies essentially on maximal vertices and finite deletion. Neither is available in general. For example, an infinite wpo may have no maximal vertex at all, so the finite construction is not a transfinite algorithm in disguise. Moreover, [proposition 8.6](#) proves that even a complete description of the abstract infinite incomparability graph cannot suffice unless its component order is retained.

A concrete remaining direction is therefore to understand the friendly ranks of *infinite connected* wpos and how they respond to finite extensions. The component decomposition reduces a general

instance to those connected pieces plus ordinal addition, but it does not calculate their ranks. We make no conjectural formula for those pieces on the basis of the finite experiment alone.

The proposed new finite characterization warrants external checking and a broader literature comparison. If it is already implicit in an earlier source, the proofs and artifacts remain useful but the novelty claim would need to be revised. This distinction is separate from the question whether the displayed theorems follow from the arguments given.

A Statement and dependency audit

The following table separates imported results from the mathematical work proved in this manuscript.

Statement	Dependency and status
Definition of friendly order type	Vialard’s invariant; not introduced here.
Multiset-width transfer	Imported theorem (9); no new proof claimed.
Ordered components	Elementary order argument, proved in lemma 3.1.
Protected-root non-cut lemma	Full finite proof in lemma 4.1.
Finite component formula	Capacity bound plus constructive lower bound; theorem 4.3.
Maximum friendly subsets	Protected-root construction and necessity argument; theorem 5.1.
Finite multiset-width formula	Component theorem plus imported width transfer.
Binary ordinal-sum law	Known law, independently proved in lemma 8.1.
Transfinite finite-block formulas	Ordinal-sum rank proof plus finite theorem.
Product and substitution laws	Direct graph arguments plus finite theorem.
Counting generating function	Unique ordered component decomposition of natural labellings.
Finite computational agreement	Executed tests within explicit bounds; not a formal proof.
Novelty/priority	Not certified by literature search or by the computation.

Checks that prevent common misstatements

The friend must be in the current residual, isolated vertices must be counted as components, and an empty base poset gives a singleton multiset order of width one. The product formula requires both factors to have at least two elements. Empty substitution fibers require deleting their base vertices first. Protected roots must be minimal *inside their component*, not necessarily globally minimal in the original poset. Forest roots for a later component may be removed when an earlier component is processed; they remain roots of the certificate forest, not necessarily members of the final residual. Finally, the outer addition in a transfinite sum is ordinal addition, not natural addition or cardinal addition.

B Archive contents and reproduction

The main files are `article.tex`, `article.pdf`, `sources.bib`, and `README.md`. Executable code is in `src/friendly.py` and `src/verify.py`; concrete inputs are in `examples/`; verification output is in `data/`. The accompanying `build.sh` compiles the article using a standard TeX installation. No font files, downloaded source papers, checksum files, or claims of external peer review are included.

The source and proofs are intentionally separable. The component formula can be checked by reading only [sections 2 to 4](#); its finite applications do not depend on transfinite notation. The ordinal consequences require [section 8](#), and the width consequences also require the explicitly imported theorem. The independent dynamic program is confined to finite computations and is not used as a premise in any proof.

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